

Regime-Conditional Reward-Free Exploration

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Abstract

We formulate reward-free exploration under hidden recurring transition regimes. Training observes an unlabelled piecewise-stationary transition stream; deployment supplies a short reward-free diagnostic phase in a previously seen mode; only then is an arbitrary reward revealed. Our Detect–Match–Explore (DME) construction quarantines boundaries, reuses returning mode models, and diagnoses deployment before planning. We prove conditional and diagnostic results but not the central detector segmentation guarantee. A ten-seed tabular pilot finds a positive but practically small DME advantage over restart, while a MiniGrid pilot favors pooling. Both predeclared gates fail.

1 Objective and relation to prior work

Stationary RFE must support every post-hoc reward [5, 6, 8]. DARLING studies online reward-bearing non-stationary RL [4]; recurring-context work reuses reward-specific policies [1]; latent MDPs optimize a known reward objective [7]; and active detection assumes known candidate models [3]. Our distinct objective jointly requires learning reward-uniform models from an unsegmented recurring stream and a reward-independent deployment diagnostic.

Let unknown episodic kernels $\{P^{(m)}\}_{m=1}^M$ share finite state/action spaces and horizon H . Hidden mode z_k is fixed within each training episode, switches only between episodes, may recur, and obeys a minimum dwell condition. No training rewards, boundaries, or labels are observed. At deployment, a previously learned mode Z stays fixed for q reward-free diagnostic episodes. Then an arbitrary pathwise-unit-bounded reward r is revealed and no further interaction is allowed. The PAC objective is

$$\Pr\left\{\sup_r[V_{P^{(Z)}}^*(r) - V_{\hat{P}^{(Z)}}(r)] \leq \epsilon\right\} \geq 1 - \delta.$$

2 Detect–Match–Explore

DME executes reward-free probes, detects a transition change, quarantines an uncertain boundary, and compares a confirmed segment with stored transition models. A separated match resumes an existing mode-specific RFE learner; otherwise DME creates a new entry. Deployment uses only a reward-free transition prefix to match or reject before the reward is known. Assumptions cover probe reachability, controlled transcript separation, clean occupancy, dwell time, transition-only changes, and adaptively valid confidence sets.

3 Established theory

Let SEG mean that accepted samples are assigned to their correct mode up to permutation, returning modes are matched, and quarantined samples never contaminate an RFE learner.

Theorem 1. *If SEG fails with probability δ_{seg} , each of M stationary RFE calls has uniform error η and failure δ_{rfe} , and deployment diagnosis fails with probability δ_{id} , then DME has uniform error η with probability at least $1 - \delta_{\text{seg}} - M\delta_{\text{rfe}} - \delta_{\text{id}}$.*

Proof. Intersect the stated success events and union bound. $\square$

Proposition 1. *For known models and fixed diagnostic strategy σ , maximum likelihood obeys $\Pr_m(\hat{m} \neq m) \leq \sum_{\ell \neq m} \text{Aff}(\mathbb{P}_m^\sigma, \mathbb{P}_\ell^\sigma)$.*

Theorem 2. *With zero deployment observations, two exactly known modes can force worst-case loss at least $1/2$.*

Proof. Reverse the good successors of two initial actions between the modes; choosing one action with probability p yields losses p and $1 - p$. $\square$

Theorem 3. *For two modes and any diagnostic test,*

$$\Pr_m(\psi = \ell) + \Pr_\ell(\psi = m) \geq \frac{1}{2} e^{-D_{\text{KL}}(\mathbb{P}_m^\sigma \| \mathbb{P}_\ell^\sigma)}.$$

This follows from binary testing and Bretagnolle–Huber [2]. A proof that the implemented detector establishes SEG and an adaptive M -mode direct-sum RFE lower bound remain open; neither is claimed.

4 Measured pilots

The finite-horizon tabular study uses ten seeds, two/three recurring modes, two separations, and post-hoc target/adversarial rewards. Values are in $[0, 1]$. DME’s paired improvement over restart is 0.00110 with 95%

Table 1: Tabular pilot; lower gap is better.

Method	Units	Accept rate	Mean worst-reward gap
Pooled	40	1.000000	0.134720962030
Restart, no reuse	40	1.000000	0.002482741316
Cluster, no quarantine	40	1.000000	0.001268908213
Recurrence-aware	40	1.000000	0.001378200021
Oracle boundary	40	1.000000	0.001627412643
Oracle mode	40	1.000000	0.001566496241
Sliding window	40	1.000000	0.154578714875

seed-bootstrap interval $[0.00057, 0.00178]$, below the predeclared 0.01 margin. Mean recurrence saving is 48.1 transitions. The gate fails.

The ten-seed symbolic MiniGrid pilot also fails: DME beats restart (0.0492 versus 0.2648 worst-task gap) but pooling obtains zero gap; deployment-ID error is 0.20.

Table 2: MiniGrid pilot.

Method	Seeds	Mean worst-task gap
pooled	10	0.000000
restart	10	0.264844
recurrence_aware	10	0.049219
oracle_upper_bound	10	0.000000

5 Limitations and conclusion

The key segmentation theorem is open, the base collector is not a faithful named minimax RFE implementation, empirical reward families are finite, and MiniGrid is too easy for pooling. Misdiagnosis can select an unsafe policy, so abstention is necessary. Regime-conditional RFE is a distinct and testable objective, but current theory and both failed gates preclude a NeurIPS-readiness claim.

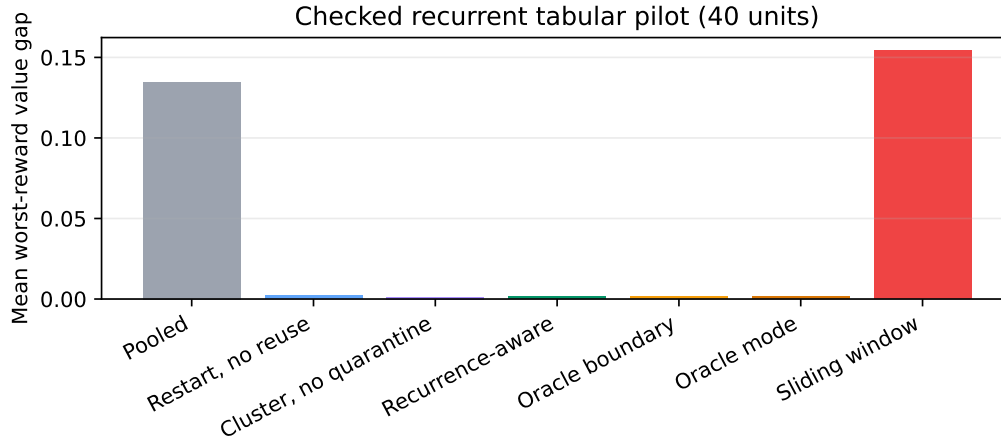


Figure 1: Tabular worst-reward gaps generated from raw results.

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